Chapter 3. Stratified Random Sampling

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1 Introduction

Definition 3.1

A **stratified random sample** splits the population into non-overlapping **strata** and draws independent SRSs within them.

Each stratum is a separate population. Stratification pays when the variation *between* strata is large: SRS models $y_i = \mu + \varepsilon_i$, stratified sampling $y_{ij} = \mu_j + \varepsilon_{ij}$. Design questions:

- how many strata to construct,
- how to allocate the total sample size to the strata,
- how to choose the strata.

2 Notation

Strata $j = 1, \ldots, L$, units i, observations y_{ij}; SRS without replacement within strata.

N_j, n_j	population and sample size of stratum j; $N = \sum N_j$, $n = \sum n_j$
$w_j = N_j/N$	stratum weight
$f_j = n_j/N_j$	sampling fraction in stratum j
$\mu_j = \frac{1}{N_j}\sum_{i=1}^{N_j} Y_{ij}$	true stratum mean
$\bar{y}_j = \frac{1}{n_j}\sum_{i=1}^{n_j} y_{ij}$	sample stratum mean
$\sigma_j^2 = \frac{1}{N_j}\sum_i (Y_{ij}-\mu_j)^2$	true stratum variance
$S_j^2 = \frac{N_j}{N_j-1}\sigma_j^2$	stratum variance with divisor $N_j - 1$
$s_j^2 = \frac{1}{n_j-1}\sum_{i=1}^{n_j}(y_{ij}-\bar{y}_j)^2$	sample stratum variance

3 Estimation of the Population Mean

Motivation. Sampling each group separately removes between-group variation; under proportional allocation it never loses to SRS.

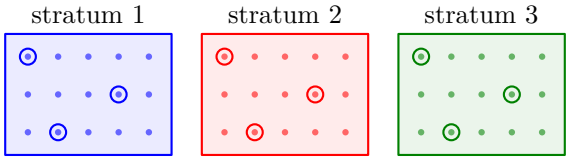


Figure. *Independent SRSs (circled) within homogeneous strata; only within-stratum variation enters* $\mathrm{Var}(\bar{y}_{st})$, $\bar{y}_{st} = \sum_h W_h \bar{y}_h$.

Theorem 3.2: Stratified estimator of μ

$$\bar{y}_{\mathrm{st}} = \frac{1}{N}\sum_{j=1}^{L} N_j \bar{y}_j = \sum_{j=1}^{L} w_j \bar{y}_j$$

is unbiased for μ, with

$$\mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{S_j^2}{n_j}, \qquad \widehat{\mathrm{Var}}(\bar{y}_{\mathrm{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{s_j^2}{n_j}, \qquad \mathrm{E}[\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{st}})] = \mathrm{Var}(\bar{y}_{\mathrm{st}}).$$

A $100(1-\alpha)\%$ interval is $\bar{y}_{\mathrm{st}} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{st}})}$.

Proof. Each $\bar{y}_j$ is an SRS mean, so by Ch. 2, $\mathrm{E}(\bar{y}_j) = \mu_j$, $\mathrm{Var}(\bar{y}_j) = (1-f_j)S_j^2/n_j$ and $\mathrm{E}(s_j^2) = S_j^2$. Thus

$$\mathrm{E}(\bar{y}_{\mathrm{st}}) = \sum_j w_j \mu_j = \frac{1}{N}(N_1\mu_1 + \cdots + N_L\mu_L) = \frac{1}{N}(\tau_1 + \cdots + \tau_L) = \frac{\tau}{N} = \mu.$$

Since the strata are sampled independently the covariances vanish and $\mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_j w_j^2 \mathrm{Var}(\bar{y}_j)$. Unbiasedness of $\widehat{\mathrm{Var}}$ follows from $\mathrm{E}(s_j^2) = S_j^2$. ■

4 Estimation of the Population Total

Motivation. Stratum totals add, and so do their variances (independent samples).

Theorem 3.3

$\hat{\tau}_{\text{st}} = N\bar{y}_{\text{st}} = \sum_{j=1}^{L} N_j \bar{y}_j$ is unbiased for τ, and

$$\text{Var}(\hat{\tau}_{\text{st}}) = N^2 \text{Var}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} N_j^2 (1 - f_j) \frac{S_j^2}{n_j}, \qquad \widehat{\text{Var}}(\hat{\tau}_{\text{st}}) = \sum_{j=1}^{L} N_j^2 (1 - f_j) \frac{s_j^2}{n_j}.$$

Example 3.4: TV-viewing time

	N_j	w_j	n_j	$\bar{y}_j$	s_j
Town A	155	0.5	20	33.90	5.95
Town B	62	0.2	8	25.12	15.25
Rural	93	0.3	12	19.00	9.36
Total	310		40		

Mean.

$$\bar{y}_{\text{st}} = \frac{155(33.90) + 62(25.12) + 93(19.00)}{310} = \frac{8578.94}{310} = 27.67.$$

$$\begin{aligned}\widehat{\text{Var}}(\bar{y}_{\text{st}}) &= (0.5)^2 \frac{5.95^2}{20}\left(1 - \frac{20}{155}\right) + (0.2)^2 \frac{15.25^2}{8}\left(1 - \frac{8}{62}\right) + (0.3)^2 \frac{9.36^2}{12}\left(1 - \frac{12}{93}\right) \\ &= 0.385 + 1.013 + 0.572 = 1.97,\end{aligned}$$

so the estimate with a 2-SD bound is $27.67 \pm 2\sqrt{1.97} = 27.67 \pm 2.81$.
Total. $\hat{\tau}_{\text{st}} = 310(27.67) = 8579$, $\widehat{\text{Var}}(\hat{\tau}_{\text{st}}) = 310^2(1.97) \approx 189{,}400$, and the bound is $2\sqrt{189{,}400} \approx 870$.

5 Sample Size for Estimating μ and τ

Setting $2\sqrt{\text{Var}(\bar{y}_{\text{st}})} = B$ alone cannot be solved for n; we must also fix how n is split among the strata. Write the allocation as $n_j = n a_j$ with $\sum a_j = 1$.

Theorem 3.5: Sample size for a given allocation

$$n = \frac{\sum_{j=1}^{L} N_j^2 S_j^2 / a_j}{N^2 D + \sum_{j=1}^{L} N_j S_j^2}, \qquad D = \begin{cases} B^2/4 & \text{for } \mu, \\ B^2/(4N^2) & \text{for } \tau, \end{cases}$$

with S_j^2 replaced by prior estimates s_j^2 in practice.

Derivation

For μ let $D = (B/2)^2 = \text{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}(1 - \frac{n_j}{N_j})$. Multiply by N^2 and substitute $n_j = n a_j$:

$$N^2 D = \sum_j N_j^2 \frac{S_j^2}{n a_j} - \sum_j N_j S_j^2 \implies n\Big(N^2 D + \sum_j N_j S_j^2\Big) = \sum_j \frac{N_j^2 S_j^2}{a_j}.$$

For τ, $\text{Var}(\hat{\tau}_{\text{st}}) = N^2 \text{Var}(\bar{y}_{\text{st}}) = B^2/4$ gives the same formula with $D = B^2/(4N^2)$. (Using 1.96 instead of 2 replaces $D = (B/2)^2$ by $(B/1.96)^2$.)

Example 3.6: Equal allocation

For the TV population $(N_j = 155, 62, 93)$ with prior variances $s_1^2 = 25$, $s_2^2 = 225$, $s_3^2 = 100$, equal allocation $a_j = 1/3$ and $B = 2$ for μ $(D = 1)$:

$$n = \frac{3(155^2 \cdot 25 + 62^2 \cdot 225 + 93^2 \cdot 100)}{310^2(1) + (155 \cdot 25 + 62 \cdot 225 + 93 \cdot 100)} = \frac{6{,}991{,}275}{123{,}225} = 56.7,$$

so $n = 57$ and $n_1 = n_2 = n_3 = 19$.

6 Allocation of the Sample

Let the survey cost be

$$TC = c_0 + \sum_{j=1}^{L} c_j n_j,$$

where c_0 is the overhead cost and c_j the cost per observation in stratum j.

6.1 Equal allocation

$n_j = n/L$. Then $TC = c_0 + \frac{n}{L}\sum c_j$, so $n = \dfrac{(TC - c_0)L}{\sum_j c_j}$.

6.2 Proportional allocation

$n_j = nw_j = nN_j/N$, i.e. $a_j = w_j$ and all $f_j = f$ are equal. Then $TC = c_0 + n\sum c_j w_j$, so $n = \dfrac{TC - c_0}{\sum_j c_j w_j}$; if $c_j \equiv c$, $n = (TC - c_0)/c$. The estimator becomes self-weighting: $\bar{y}_{\text{st}}$ equals the plain mean of all n observations.

6.3 Optimum allocation

Theorem 3.7: Optimum allocation

For a fixed total cost (equivalently, for a fixed variance) $\text{Var}(\bar{y}_{\text{st}})$ is minimized when

$$n_j \propto \frac{w_j S_j}{\sqrt{c_j}}, \qquad \text{i.e.} \qquad n_j = n\,\frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}},$$

and the total sample size under a budget TC is

$$n = \frac{(TC - c_0)\sum_k w_k S_k/\sqrt{c_k}}{\sum_k w_k S_k \sqrt{c_k}}.$$

Derivation: Cauchy–Schwarz argument

Split the variance into the part depending on the allocation and a constant:

$$\text{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}\left(1 - \frac{n_j}{N_j}\right) = \underbrace{\sum_j \frac{w_j^2 S_j^2}{n_j}}_{V'} - \underbrace{\sum_j \frac{w_j S_j^2}{N}}_{\text{const}}.$$

Let $C' = TC - c_0 = \sum_j c_j n_j$. Minimizing V' for fixed C', or C' for fixed V', is the same as minimizing the product

$$V'C' = \Big(\sum_j \frac{w_j^2 S_j^2}{n_j}\Big)\Big(\sum_j c_j n_j\Big) = \Big(\sum_j a_j^2\Big)\Big(\sum_j b_j^2\Big), \quad a_j = \frac{w_j S_j}{\sqrt{n_j}},\ b_j = \sqrt{c_j n_j}.$$

By the Cauchy–Schwarz inequality

$$V'C' \geq \Big(\sum_j a_j b_j\Big)^2 = \Big(\sum_j w_j S_j \sqrt{c_j}\Big)^2,$$

whose right-hand side does not depend on the n_j. Equality holds iff b_j/a_j is constant:

$$\frac{b_j}{a_j} = \frac{n_j \sqrt{c_j}}{w_j S_j} = \text{const} \iff n_j \propto \frac{w_j S_j}{\sqrt{c_j}}.$$

Substituting $n_j = n \frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}}$ into $TC = c_0 + \sum c_j n_j$ gives $TC = c_0 + n\alpha$ with $\alpha = \frac{\sum_k w_k S_k \sqrt{c_k}}{\sum_k w_k S_k/\sqrt{c_k}}$, so $n = (TC - c_0)/\alpha$.

We therefore take a larger sample in a stratum that is (1) larger, (2) more variable, or (3) cheaper to sample.

6.4 Neyman allocation

Theorem 3.8: Neyman allocation

If the costs are equal, $c_j \equiv c$, the optimum allocation becomes

$$n_j = n \frac{w_j S_j}{\sum_k w_k S_k} = n \frac{N_j S_j}{\sum_k N_k S_k}.$$

Example 3.9: Neyman allocation for the TV data

With $n = 40$ and $N_j s_j = 922.3,\ 945.5,\ 870.5$ (sum 2738.2), the Neyman allocation is

$$n_1 = 13.5, \quad n_2 = 13.8, \quad n_3 = 12.7 \ \Rightarrow\ (13, 14, 13),$$

versus the proportional $(20, 8, 12)$ used: the small, very variable Town B gets many more. With s_j for S_j, $\mathrm{Var}(\bar{y}_{\mathrm{st}})$ drops from 1.97 to about 1.66.

7 Estimation of a Population Proportion

Motivation. Proportions are means of 0–1 variables, with $p_h(1 - p_h)$ as stratum variances.

With $y_{ij} \in \{0, 1\}$ the stratum mean $\bar{y}_j$ is the stratum proportion $\hat{p}_j$, and

$$\hat{p}_{\mathrm{st}} = \frac{1}{N}(N_1 \hat{p}_1 + \cdots + N_L \hat{p}_L) = \sum_j w_j \hat{p}_j, \qquad \mathrm{E}(\hat{p}_{\mathrm{st}}) = \sum_j w_j p_j = p.$$

Theorem 3.10

$$\mathrm{Var}(\hat{p}_{\mathrm{st}}) = \sum_j w_j^2 \frac{p_j q_j}{n_j}\Big(\frac{N_j - n_j}{N_j - 1}\Big), \qquad \widehat{\mathrm{Var}}(\hat{p}_{\mathrm{st}}) = \sum_j w_j^2 \frac{\hat{p}_j \hat{q}_j}{n_j - 1}(1 - f_j),$$

and a $100(1-\alpha)\%$ interval for p is $\hat{p}_{\mathrm{st}} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{p}_{\mathrm{st}})}$.

These follow from Ch. 2 applied stratum by stratum, the divisor $n_j - 1$ making each term unbiased.

8 Selected exercises

Motivation. Exact gains of Neyman over proportional allocation and of proportional over SRS, and a design with unequal costs.

Exercise 3.1 (Proportional versus Neyman allocation)

With equal costs and fixed n, let $\bar{y}_{\mathrm{prop}}$ and $\bar{y}_{\mathrm{opt}}$ be the stratified means under $n_j = n w_j$ and Neyman $n_j = n w_j S_j / \bar{S}$, $\bar{S} = \sum_j w_j S_j$ ($n_j \le N_j$). Show that

$$\mathrm{Var}(\bar{y}_{\mathrm{prop}}) - \mathrm{Var}(\bar{y}_{\mathrm{opt}}) = \frac{1}{n}\sum_{j=1}^{L} w_j (S_j - \bar{S})^2,$$

so $\mathrm{Var}(\bar{y}_{\mathrm{prop}}) > \mathrm{Var}(\bar{y}_{\mathrm{opt}})$ unless all S_j are equal.

Solution.

Step 1. (A common constant.) For any allocation,

$$\mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}\Big(1 - \frac{n_j}{N_j}\Big) = \sum_j \frac{w_j^2 S_j^2}{n_j} - \frac{1}{N}\sum_j w_j S_j^2,$$

since $w_j^2/N_j = w_j/N$. The last sum does not depend on the allocation.

Step 2. (Proportional.) With $n_j = n w_j$, $\sum_j w_j^2 S_j^2/n_j = \frac{1}{n}\sum_j w_j S_j^2$.

Step 3. (Neyman.) With $n_j = n w_j S_j/\bar{S}$, $\sum_j w_j^2 S_j^2/n_j = \frac{\bar{S}}{n}\sum_j w_j S_j = \frac{\bar{S}^2}{n}$.

Step 4. (Difference.) Subtracting, the constants of Step 1 cancel and

$$\mathrm{Var}(\bar{y}_{\mathrm{prop}}) - \mathrm{Var}(\bar{y}_{\mathrm{opt}}) = \frac{1}{n}\Big(\sum_j w_j S_j^2 - \bar{S}^2\Big) = \frac{1}{n}\sum_j w_j (S_j - \bar{S})^2,$$

the last step being the identity $\sum_j w_j (S_j - \bar{S})^2 = \sum_j w_j S_j^2 - 2\bar{S}\sum_j w_j S_j + \bar{S}^2 \sum_j w_j = \sum_j w_j S_j^2 - \bar{S}^2$ (recall $\sum_j w_j = 1$).

Step 5. The right side is a positively weighted variance of the S_j, zero iff all $S_j = \bar{S}$ (then the allocations coincide): Neyman gains when strata differ in variability. ■

Exercise 3.2 (Proportional allocation versus SRS)

Let $\bar{y}_{\text{ran}}$ be the mean of an SRS of size n from the whole population, with variance $\frac{1-f}{n}S^2$. Show that

$$\operatorname{Var}(\bar{y}_{\text{ran}}) - \operatorname{Var}(\bar{y}_{\text{prop}}) = \frac{1-f}{n(N-1)}\Big[\sum_{j=1}^{L} N_j(\mu_j-\mu)^2 - \sum_{j=1}^{L}(1-w_j)S_j^2\Big],$$

and conclude that proportional allocation is better than SRS unless the stratum means are nearly equal.

Solution. [Supplement]

Step 1. (Analysis of variance of the population.) Adding and subtracting μ_j inside stratum j, the cross term vanishes because $\sum_i (Y_{ij}-\mu_j) = 0$:

$$(N-1)S^2 = \sum_j \sum_i (Y_{ij}-\mu)^2 = \sum_j (N_j-1)S_j^2 + \sum_j N_j(\mu_j-\mu)^2.$$

Step 2. Under proportional allocation $f_j = f$, so $\operatorname{Var}(\bar{y}_{\text{prop}}) = \sum_j w_j^2(1-f)\frac{S_j^2}{nw_j} = \frac{1-f}{n}\sum_j w_j S_j^2$. Hence

$$\operatorname{Var}(\bar{y}_{\text{ran}}) - \operatorname{Var}(\bar{y}_{\text{prop}}) = \frac{1-f}{n}\Big(S^2 - \sum_j w_j S_j^2\Big).$$

Step 3. By Step 1,

$$S^2 - \sum_j w_j S_j^2 = \frac{\sum_j N_j(\mu_j-\mu)^2}{N-1} + \sum_j S_j^2\Big(\frac{N_j-1}{N-1} - \frac{N_j}{N}\Big),$$

and $\frac{N_j-1}{N-1} - \frac{N_j}{N} = \frac{N(N_j-1)-N_j(N-1)}{N(N-1)} = -\frac{N-N_j}{N(N-1)} = -\frac{1-w_j}{N-1}$. Substituting gives the claim.

Step 4. The first sum is the between-strata sum of squares, growing with N_j; the second sums L within-stratum variances with coefficients below 1. Unless the μ_j are nearly equal, $\operatorname{Var}(\bar{y}_{\text{prop}}) < \operatorname{Var}(\bar{y}_{\text{ran}})$; for large N_j the difference is about $\frac{1-f}{n}\sum_j w_j(\mu_j-\mu)^2 \geq 0$. ■

Exercise 3.3 (Allocation with unequal costs)

A county estimates the proportion of families who will use an expanded day-care facility. Strata: city users, country users, city nonusers, country nonusers, $N_j = 97, 43, 145, 68$; expected proportions 0.9 (users) and 0.5 (nonusers); costs 4 (user) and 8 (nonuser). Find n and the allocation for $B = 0.05$.

Solution. For 0/1 data $S_j^2 = \frac{N_j}{N_j-1}p_jq_j \approx p_jq_j$, so we use $p_jq_j = 0.09, 0.09, 0.25, 0.25$ in place of S_j^2.

Step 1. (Allocation.) By the optimum-allocation theorem, $n_j \propto w_jS_j/\sqrt{c_j} \propto N_j\sqrt{p_jq_j/c_j}$:

$$N_j\sqrt{p_jq_j/c_j} = 14.55,\ 6.45,\ 25.63,\ 12.02, \qquad \text{sum } 58.65,$$

so $a_j = 0.248,\ 0.110,\ 0.437,\ 0.205$.

Step 2. (Sample size.) With $N = 353$, $D = B^2/4 = 0.000625$ and the sample-size theorem,

$$n = \frac{\sum_j N_j^2 p_jq_j/a_j}{N^2D + \sum_j N_jp_jq_j} = \frac{22{,}595}{77.88 + 65.85} = \frac{22{,}595}{143.73} = 157.2,$$

where $\sum_j N_j^2 p_j q_j / a_j = 846.8/0.248 + 166.4/0.110 + 5256.3/0.437 + 1156.0/0.205$ and $\sum_j N_j p_j q_j = 8.73 + 3.87 + 36.25 + 17.00$.

Step 3. $n = 158$ gives $n_j = 39.2,\ 17.4,\ 69.0,\ 32.4$, i.e. $(39, 17, 69, 33)$. Nonusers get more despite higher cost: $pq = 0.25$ vs 0.09, only partly offset by $1/\sqrt{c_j}$. ■